

CUNNAMULLA, Australia

WMO: 945000

Lat: 28.071S

Lon: 145.681E

Elev: 191

StdP: 99.06

Time Zone: 10.00 (E10)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.9	5.1	-7.1	2.1	21.4	-5.2	2.5	19.8	7.9	20.4	6.5	17.2	1.0	180	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	40.0	20.7	38.6	20.4	37.3	20.1	24.7	32.8	23.8	31.4	23.0	30.4	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.1	27.1	21.9	17.0	26.8	20.9	15.9	26.3	76.4	32.9	72.6	31.3	69.4	30.0	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.9	6.0	5.3	DB	1.5	42.6	1.2	1.5	0.6	43.7	0.0	44.6	-0.7	45.5	-1.5	46.6	(4)
			WB	0.2	26.9	1.3	2.5	-0.7	28.6	-1.5	30.1	-2.2	31.5	-3.1	33.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.6	29.6	28.4	26.2	21.4	16.8	13.7	13.0	14.9	19.1	22.2	25.4	28.3
	DBStd	6.66	3.28	3.31	2.99	3.27	2.88	3.15	2.71	3.42	3.98	3.41	3.95	3.21
	HDD10.0	8	0	0	0	0	0	3	4	2	0	0	0	0
	HDD18.3	544	0	0	0	10	63	143	167	117	38	6	1	0
	CDD10.0	4226	609	515	501	342	212	114	97	154	274	379	461	568
	CDD18.3	1721	351	282	243	102	17	4	1	12	61	127	211	309
	CDH23.3	21687	4936	3732	2798	978	107	26	6	141	731	1456	2554	4222
CDH26.7	11524	2983	2102	1372	351	9	0	0	31	266	647	1324	2439	
Wind	WSAvg	2.0	2.2	2.3	2.0	1.7	1.6	1.7	1.7	1.9	2.1	2.2	2.4	2.3
Precipitation	PrecAvg	384	63	40	40	22	36	20	23	20	16	26	35	35
	PrecMax	722	244	247	170	105	151	86	112	87	65	110	137	139
	PrecMin	123	1	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	137	61	45	44	30	38	21	27	18	17	25	35	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.3	40.9	38.3	33.7	28.1	25.9	24.1	30.4	34.6	36.8	40.3	40.1
		MCWB	21.0	22.0	20.4	19.8	16.6	16.5	13.9	16.8	16.7	17.7	18.2	20.0
	2%	DB	40.5	38.9	35.9	31.9	26.3	24.2	22.3	26.8	32.0	34.5	37.7	38.9
		MCWB	21.3	21.5	20.2	17.9	16.2	15.7	13.1	14.9	16.2	17.5	18.2	20.0
	5%	DB	38.9	37.3	34.4	30.4	24.7	22.2	20.9	24.6	30.1	32.7	35.5	37.6
		MCWB	21.0	21.4	20.1	17.4	15.6	14.3	12.4	13.8	15.8	17.4	18.5	19.8
	10%	DB	37.2	35.5	32.9	28.6	23.0	20.0	19.1	22.4	27.9	30.6	33.4	36.2
		MCWB	20.6	21.1	19.5	16.7	15.0	13.4	11.5	12.9	14.8	16.5	18.3	19.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.2	26.1	24.5	23.0	20.4	17.8	16.7	18.1	19.6	22.1	23.6	24.5
		MCDB	35.4	32.8	30.1	30.1	21.7	23.6	20.1	27.6	25.3	30.9	30.2	34.3
	2%	WB	24.8	24.6	23.3	20.9	18.7	16.5	14.7	16.6	17.9	20.1	22.2	23.1
		MCDB	34.0	32.1	29.1	25.7	23.1	21.7	20.2	24.3	26.5	27.6	28.6	31.9
	5%	WB	23.7	23.8	22.1	19.3	17.5	15.6	13.7	15.1	17.1	18.9	21.3	22.0
		MCDB	31.9	31.4	29.4	25.0	22.3	20.6	18.9	22.2	26.9	26.8	27.7	31.0
	10%	WB	22.8	23.1	21.2	18.1	16.4	14.4	12.5	13.9	16.1	17.9	20.5	21.2
		MCDB	31.0	29.9	29.2	25.5	21.1	19.0	17.5	20.5	25.5	26.6	27.2	31.8
Mean Daily Temperature Range	5% DB	MDBR	12.0	11.5	12.1	12.3	11.0	11.5	12.2	13.4	14.0	13.3	12.1	12.6
		MCDBR	14.0	13.8	13.6	14.3	12.9	13.0	14.0	15.4	15.7	16.3	14.3	14.2
		MCWBR	4.5	4.4	4.9	5.6	5.8	6.3	6.8	7.1	6.2	6.3	5.1	4.8
	5% WB	MCDBR	11.5	11.1	10.4	10.7	9.4	11.2	11.2	13.0	13.7	12.4	10.6	11.6
		MCWBR	4.4	4.7	4.4	5.1	4.7	6.0	6.3	6.5	5.9	6.3	4.6	4.7
Clear-Sky Solar Irradiance	taub	0.355	0.350	0.331	0.315	0.297	0.288	0.286	0.299	0.338	0.344	0.358	0.354	
	taud	2.520	2.550	2.595	2.589	2.593	2.621	2.603	2.541	2.425	2.436	2.450	2.490	
	Ebn at Noon	986	975	964	932	906	894	911	936	939	969	977	989	
	Edh at Noon	113	107	96	88	79	73	77	90	112	118	120	117	
All-Sky Solar Radiation	RadAvg	7.62	7.05	6.21	5.13	4.02	3.34	3.77	4.78	5.93	6.95	7.34	7.69	
	RadStd	0.38	0.49	0.35	0.29	0.18	0.22	0.24	0.23	0.36	0.45	0.64	0.51	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)		+0.66	N/A	N/A	+0.88	N/A	N/A	N/A	N/A	N/A	+166

Nomenclature: See separate page